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PERIODICITY OF SPAWNING AND DURATION OF INVCUBATION TIME FOR *Chionoecetes opilio* IN THE GULF OF ST. LAWRENCE.

by

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Summary

We used histology, and microscopy observations as well as simple colour indices to monitor the seasonal development of the egg stages in the brood and the development of the maturity stages of the gonads. Over a period of 21 months of monitoring, we showed that the complete sequence of development of the eggs takes two years. The gonads can fully develop in one year, but a new spawning will not take place until the previous batch hatches.

The usual habitat for snow crab, *Chionoecetes opilio*, in the Gulf of St. Lawrence, is a layer of intermediate water which remains all year round at -1 to +1°C and 32.5 to 33 ‰ salinity. Some females however move into shallower, warmer water in the spring before hatching of the eggs, mating and spawning. They may remain in warmer water for several weeks, which may enhance the development of their eggs, and a one year development may then be achieved as it is observed in aquarium.

The life span of females after the terminal molt does not normally exceed 5 years. A female will therefore produce no more than two or at most three clumps of eggs in its life span, unless it spends some time in shallow water.

Introduction

Some 20 000 tons of snow crab (*Chionoecetes opilio*) are harvested each year in the Gulf of St. Lawrence (Canada). The commercial fishery developed some 25 years ago as a market was opened by the decline in Alaskan crab landings. The landings peaked to 32 000 tons in 1982, but appear to be very dependent on year to year fluctuations in recruitment.

The reproductive biology of snow crab in the Gulf of St.Lawrence is now starting to be understood (Comeau and Conan, 1992; Conan *and* Comeau, 1986; Sainte Marie and Hazel, 1992). Males and females grow to a terminal molt which is reached over a wide range of ages and sizes by individuals within a same cohort. Males become efficiently sexually active only after reaching the terminal molt. Females mate and spawn for the first time after the terminal molt, while their shell is still soft, but normally mate and spawn again as soon as their previous batch of eggs hatches. According to aquarium observations (Moriyasu and Conan, 1988), the first spawning takes place as early as February. The process of mating immediately followed by spawning is well synchronized among terminal molt hard shell (multiparous) females. It has been observed to take place in May and June, both in the laboratory and in the wild. The sequential process of

hatching/mating/spawning appears to be triggered in early spring by detrital organic matter released from the spring plankton bloom (Comeau *et al.*,1991).

Until recently, it was believed that the eggs would be carried under the abdomen of the females for a period of one year and would develop during this time until the next spring, May or early June, when the larvae would be released in the plankton. The females were observed in the wild to mate and spawn immediately after the release of the larvae. Very few adult females are ever found carrying no eggs less than 5% (Mallet *et al.* 1992). The belief that the duration of the incubation period was no longer than one year was prompted by aquarium observations (Watson, 1970), Bernard Sainte Marie (personal communication, Ministère des Pêches et Océans, Institut Maurice Lamontagne, Mont Joli, Québec), and by observations made by Kon (1980) for *Chionoecetes opilio elongatus* in the Japan Sea.

However, it is difficult to maintain *C. opilio* in aquarium for long periods at a temperature matching the natural environment. *C. opilio* in the Gulf of St. Lawrence live on muddy bottoms flushed all year round by an intermediate layer of water at a very stable temperature of -1 to + 1º C, 33 ‰ salinity, which is believed to come from Labrador and overflow the strait of Belle Isle in winter. Surface water temperatures in the Gulf of St. Lawrence may reach 20ºC and 20‰ salinity in the summer.

Conan *et al.* (1989) suggested that the development of the eggs in the natural environment may take 24 months or more and that the first spawning of the females (primiparous females) took place several months before the spawning of the multiparous females. Their suggestion was prompted: 1) by aquarium observation of females maintained all year round at low temperature, and 2) by field observations showing that a large number of females already carried orange (recently spawned to mid developed eggs) early in the spring when the hatching, mating and spawning sequence had not yet been initiated by multiparous females carrying black (fully developed) eggs. Kanno (1987) had independently made similar observations on *Chionoecetes japonicus* from the Sea of Japan.

In order to trace the fluctuations in recruitment back to parental stocks it was important to identify the relative contribution of primiparous and multiparous females to egg production, and to demonstrate whether the brood carrying period lasted one or two years in the natural environment of the Gulf of St. Lawrence. In order to provide an objective response, we decided to follow concomitantly the stages of development of both the eggs and the gonads all year round in samples taken at regular time intervals. The definition of the development stage of the eggs and of the gonads needed to be defined according to more rigorous criteria than a subjective reading of colour, histological techniques were used.

Material and methods

Sea sampling

Seasonal coverage

The females were collected in Baie des Chaleurs (47º 55' N, 64º 35' W) at depths varying from 80 to 90 m using a Bay of Biscay Bigouden *Nephrops* trawl hauled by 18.5m research ship Opilio. From May 1989 to December 1990, 54 trawl hauls provided a set of samples totalling 629 females.

Cursory information was recorded for each female captured: state of fouling of the carapace (clean, presence of epibionts, full cover of epibionts), colour of the eggs (orange, brown, black), colour of the gonad (beige, light orange, bright orange), width of carapace, reproduction instances: either

primiparous or multiparous. Primiparous females were identified as clean carapace females bearing no grasping marks. The primiparous females have mated shortly after the molt to morphometric maturity, while still in soft carapace state, they carry their first brood. Males hold tightly the pereiopods of the females in their claws during the precopulatory embrace and conspicuous scars can be observed after mating, except if the carapace of the female was still soft at the time of mating, i.e. except if the female is primiparous.

A subsample was taken every month for detailed analysis. Five females were selected within each available 5 mm size class and colour of eggs. A gonad, and about a hundred eggs were taken from each female and preserved in Bouin fluide for histology or direct light microscopy observation.

Geographic coverage
An additional sample of 9 586 females was collected over the 144 stations distributed on the whole extent of the Gulf of St. lawrence fishery between July 10 and November 7 1991. Only cursory information was recorded.

Laboratory processing

Cursory observations based on colour of the eggs or of the gonads can be made at sea on a large number of individuals. However, these simple indices may be unreliable and needed to be calibrated by comparison with a subsample studied in detail at the laboratory by histology and microscopic observations.

Ovocyte development and gonad colour
The ovaries were fixed in Bouin's fluid for 48 hours, and then preserved in 70% ethanol. Standard techniques were used for inclusion in paraffin wax. The tissues were then sectioned at 7 µm in thickness and stained by Masson's trichrome, using picrofuchsine as an intermediate stain. The preparations were observed by standard light microscopy at 60X to 400X magnifications and photographed. The stages of development of the ovocytes were identified according to the criteria described by Laulier (1974), Johnson (1980) and Kon (1980)

Embryo development and egg colour
The eggs were preserved in Bouin's fluid during 72 hours and observed "in toto" under a stereomicroscope at magnifications varying from 16X to 40X. The stages of development were defined as described by Kon (1980). A few drops of light green at 1% dilution were added to increase contrast and improve observation of the appendages.

Graphics analysis

Simple pie charts, histograms and scatter plots allow to visualize the relationship between colour and stage of development of the eggs, colour and stage of development of the gonad, and seasonal variations of the proportions in each category defined.

Results and discussion

The list of samples collected is summarized in table 1. The series of observations covers two consecutive spring periods. Samples could not be collected during the months of February, March, and April due to the thick ice cover during these months.

Subsamples were collected for detailed light microscopy analysis during the months of January, June and September. Six stages of gonad development were observed by histology, and

regrouped into 3 conspicuous categories: previtellogenesis, early vitellogenesis, advanced vitellogenesis (Fig 1). There is a good correlation between gonad stages and cursory observations based on external colour (Fig. 2), although histology observations allow to distinguish two consecutive categories in the beige and light orange colours. The colour of the gonad can be used as a rough index of gonad development for cursory statistical observations, but not for precise determinations.

Six stages of development of the embryos could be identified in eggs incubated by the females (Fig 3). There is a good concordance between the colour of the eggs and the stages of development, however cursory observations based on the colour of the eggs allow for the discrimination of only two conspicuous categories. The distinction between several grades of "orange" and "brown" or "black" is subjective (Fig. 4). There is a wide overlap between "orange" and "brown-black". Both colour categories may regroup eggs at considerably different stages of development, and cursory observations based on egg colour only, cannot be considered reliable.

The results of the cursory analysis of the samples of primiparous females taken *in toto* are presented in figures 5a. In March, immediately before the hatching-spawning season, primiparous females carry either black eggs with orange gonad (eggs about to hatch and gonad about to spawn) or brown eggs with light orange gonad (eggs not yet ready to hatch, gonad not yet ready to spawn). The category orange eggs/gonad beige indicating a very recent spawning and is present only in May and June. The category orange eggs/light orange gonad indicating recent spawning start to appear only in May, becomes abundant in June, less abundant in July and has disappeared in August. The category brown eggs/light orange gonad appears to progressively replace the two former categories, it is the only one found in August. A brown egg/orange gonad appears in August or September and is mostly predominant until January but coexists with the previous category. Black eggs/orange gonad are found only in March when they appear to replace the previous category. Since the category brown eggs/light orange gonad which corresponds to females in an intermediate stage, is found all year round, it is tempting to conclude that the maturation of the gonad and of the eggs takes two years or more for at least some of the females. If a complete reproductive cycle was accomplished for all females in one year, since egg hatching appears is well synchronized and occurs in the spring, no brown egg females with light orange gonad (intermediate stage of development of the eggs and of the gonad), should be found in the spring and early summer

The results of the cursory analysis of the samples of multiparous females taken *in toto* are presented in figure 5b. The interpretation is complicated by the fact that there are many more categories than for primiparous females. A small proportion of females bore black, ready to hatch eggs, but light orange gonad not ready to spawn in April-May; these were likely to be senescent females since healthy females are never found without brood. The category orange eggs/orange gonad, intermediate stage of development is found without discontinuity from September 89 to August 90, through the hatching-mating-spawning period of March, April, May 90. It is logical to conclude that at least part of the females went through a two year cycle of gonad and egg maturation. From May 89 to January 90 the category occupying an equivalent position in the samples was brown eggs/orange gonad, while this category started to appear only in August in 1990 from the orange egg/orange gonad category. In 1989 the development of the eggs was more advanced than in 1990, as confirmed by the presence of black eggs as early as March. Otherwise the coexistence of two groups of females with distinct stages of egg and gonad development can be followed through 21 months surveyed and confirms the existence of a two year cycle.

The results of the detailed analysis by histology and microscopic observations are presented in table 2. Clean carapace females are thought to be primiparous. Medium fouled carapace females

are thought to be females in their second year after terminal molt, the data analysis should reveal whether these females are primiparous or multiparous. Females with fouled carapaces are multiparous, they bear conspicuous grasping marks, and some of them may be senile.

Percent frequencies of the presence of the stages in June, September, and January are presented as histograms in figures 6a,b. The percent frequencies of occurrence of the colours of the gonads and eggs are presented as histograms for comparison in figures 7a,b. Colour does not appear to be a sufficiently precise criterium for identification of the stage of development of the gonads or eggs of individual females, it has only be used as an index for statistics on large numbers. The range of development stages of light orange gonads (0 to 4) overlaps with beige gonads (0-1) and bright orange gonads (2 to 6). The range of development of brown eggs (2 to 6) overlaps with orange eggs (0 to 4). Black eggs are likely to represent only the latest of stage 6, shortly prior to hatching.

The evolution of the egg development stages can be interpreted as such in figure 6a. The stages 0-1 are found in June clean carapace (primiparous) females shortly after spawning, these eggs develop into stages 2-3-4 in September, into stages 2+ in January, into 4 over the next June. By this time the primiparous females have medium fouled carapaces. Stages 3 were found in September, 5 in January. The presence of stage 0-1 eggs in June for medium fouled females may be due to a problem of misclassification. The criterium of cleanliness of the carapace is not fully reliable, and some females might actually have been multiparous. Among females bearing fouled carapaces, taken as multiparous, two groups of egg development can be interpreted as 0-1 and 3-4 in June (shortly after egg hatching, no black eggs were present), 1 and 4-6 in September, 0-2 and 5-6 in January.

The evolution of the gonad stages in figure 6b is less informative, gonads appear to have a wide range of development stages among different individuals of similar egg stages. Some gonads may mature over one year from stages 0 to 4 in June to stages 2-6 in January for the primiparous females. By the next January (primiparous with medium fouled carapaces) 50% are already in stages 5-6. Once they reach maturity gonads seem to stay in the wait until the hatching of the eggs of the previous brood takes place. The changes in the stages of the gonads of multiparous females are not informative, a wide range of stages encountered over any of the periods studied.

Histology of the gonad and observation of the eggs under microscope provides reliable means for reading development stages of the gonad and of the eggs, but does not allow to process a number of samples large enough to reliably follow the seasonal evolution of the stages in the population. Colour can efficient be used as an index of development stage for processing large samples. Histology and observation under microscope can efficiently be used for the identification of subsamples in doubtful cases or for individual determinations..

Conclusions

The presence in the wild of females bearing medium developed orange egg, shortly before the hatching-mating-spawning period of the multiparous females in May June, can be satisfactorily explained by a two year development cycle of the eggs. The alternative hypothesis that primiparous females spawned earlier in the year than the multiparous females and appeared in the catch bearing orange eggs before the hatching-mating-spawning period is not sufficient for interpreting the data. The eggs of the females either primiparous or multiparous, develop into larvae over a period of two years.

The development of the gonads can take place over a period shorter than two years and eventually over one year. It is the development of the eggs into larvae which slows down the reproductive

cycle.

Our observations are valid only for locations of the Gulf of St. Lawrence where females remain in the intermediate water layer (50-200m) at -1 to +1ºC all year round. In certain locations such as Bonne Bay fjord or Baie St Marie, where the depth gradient is steep, it has been observed that mating couples would congregate in shallow water in the spring (Hooper, 1986; Comeau *et al, 1991*; Sainte Marie and Hazel, 1992). The couples move down to deeper water as soon as a warmer layer of surface water (10-15ºC) starts to form (Conan, personal observations). However, Conan observed by tagging females in couples with infrasonic transmitters, that some of the recently spawned females would remain in shallow water for several weeks before moving back to the intermediate layer. Aquarium observations (Moriyasu, personal observations) show that broods can develop in one year if the females are kept in water at 3 to 6ºC, although the metabolic expenses of the female are much greater. The extended stay in shallow water may be a strategy for shortening the development cycle of the eggs from two to one year.

Conan, Brichet[1], Reyss[1] and Lalou[1] (personal communication) have shown by radioisotope dating of the carapace, that females do not normally survive more than five years after the terminal molt. Unless they spend sometime in shallow water, the females will produce no more than two batches of eggs during their whole life span.

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1 Centre des Faibles Radio Activités, Laboratoire Mixte CNRS CEA, Domaine du CNRS, Gif sur Yvette, FRANCE.

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Table 1 Number of females sampled each month between May 1989 and December 1990.

Month	# of females sampled	
	Primiparous	Multiparous
May, 1989	21	27
June, 1989	2	13
July, 1989	0	15
August, 1989	8	15
September, 1989	18	13
November, 1989	5	11
December, 1989	14	16
January, 1990	20	10
March, 1990	2	10
April, 1990	0	10
May, 1990	3	35
June, 1990	67	28
July, 1990	28	28
August, 1990	32	2
September, 1990	53	45
October, 1990	59	21
November, 1990	23	21
December, 1990	5	10

Table 2 Results of the sub samples studied in detail at the laboratory

Females with new clean carapace no grasping mark (primiparous)

Month	Number of females	Gonads colour			Gonads stage			Eggs colour			Eggs stage						
		1	2	3	0-1	2-3-4	5-6	1	2	3	0	1	2	3	4	5	6
January	4	0	0	4	0	3	1	4	0	0	0	0	4	0	0	0	0
June	5	0	5	0	2	3	0	5	0	0	4	1	0	0	0	0	0
September	5	0	0	5	1	4	0	0	5	0	0	0	1	3	1	0	0

Females with intermediate carapace, presence of some epibionts and presence of grasping marks

Month	Number of females	Gonads colour			Gonads stage			Eggs colour			Eggs stage						
		1	2	3	0-1	2-3-4	5-6	1	2	3	0	1	2	3	4	5	6
January	4	0	0	4	0	2	2	0	4	0	0	0	0	0	0	4	0
June	13	6	0	7	6	6	1	13	0	0	1	6	0	0	5	0	0
September	5	0	0	5	0	0	5	0	5	0	0	0	0	5	0	0	0

Females with old fouled carapace and grasping marks present (multiparous)

Month	Number of females	Gonads colour			Gonads stage			Eggs colour			Eggs stage						
		1	2	3	0-1	2-3-4	5-6	1	2	3	0	1	2	3	4	5	6
January	12	0	0	12	0	5	7	6	6	0	1	0	4	0	0	5	2
June	17	11	0	6	10	4	3	16	1	0	3	8	0	1	5	0	0
September	6	0	0	6	1	0	5	1	5	0	0	1	0	0	4	0	1

Gonads colour:

1 - beige
2 - light orange
3 - orange

Gonads stage:

0 - 1 previtellogenesis
2 - 3 - 4 early vitellogenesis
5 - 6 advanced vitellogenesis

Eggs colour:

1 - orange
2 - brown
3 - black

Eggs stage:

0 - vitellus
1 - blastula
2 - gastrula
3 - nauplius
4 - meta-nauplius
5 - pigment stage
6 - chromatophors formation

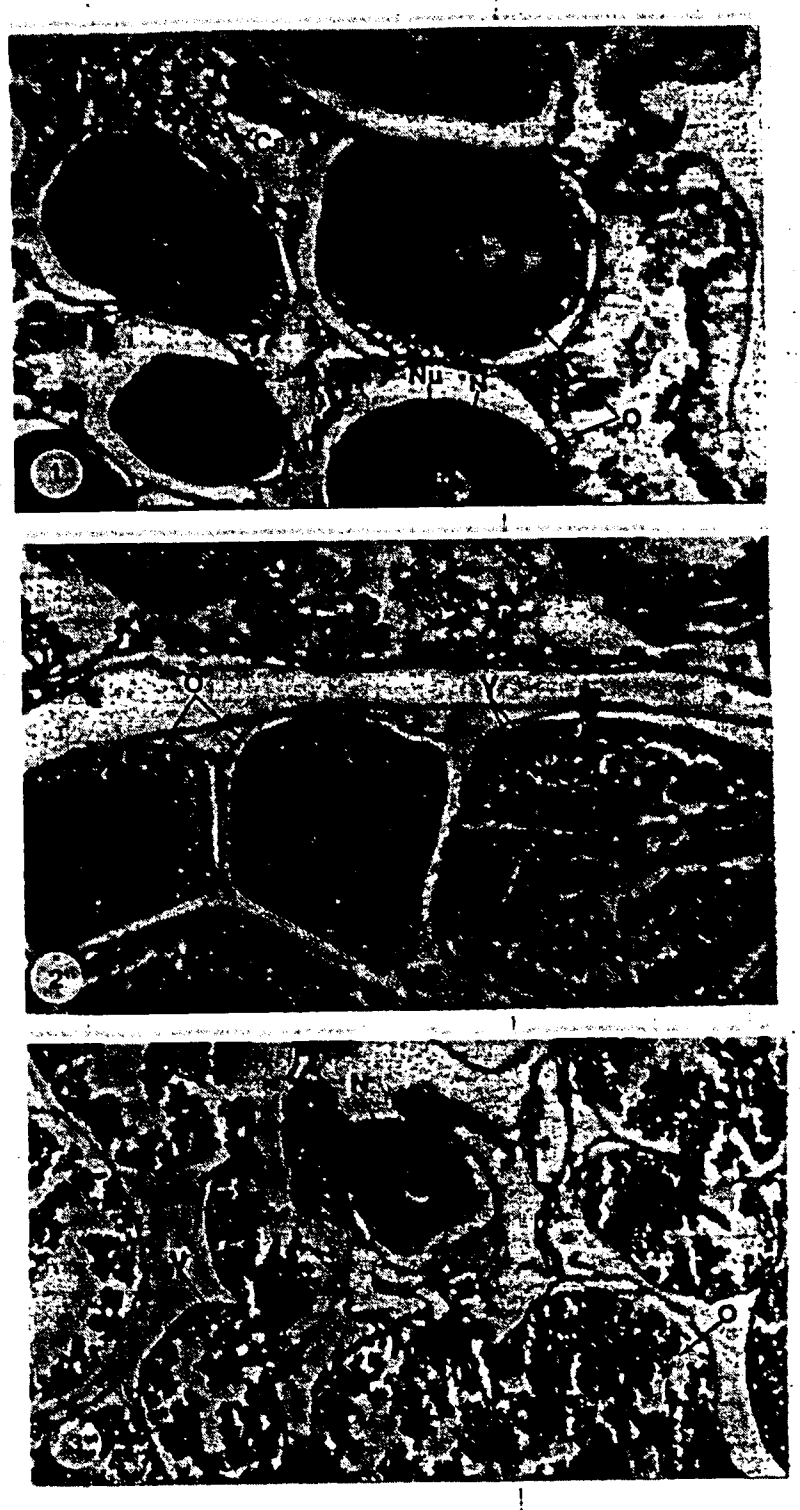


Figure 1 Gonad stages observed by histology: 1- previtellogenesis (100X), 2- early vitellogenesis (100X), 3- advanced vitellogenesis (64X).

Ca= accesory cells, N= nucleus, Nu=nucleolus, O= oocytes, Y=yolk

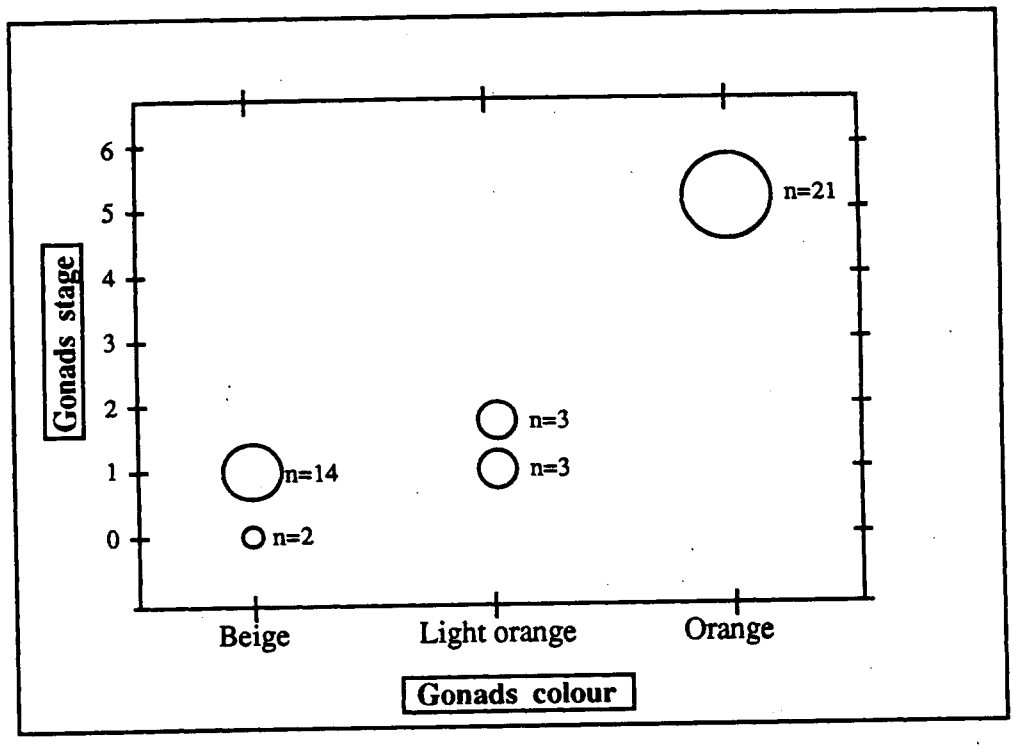
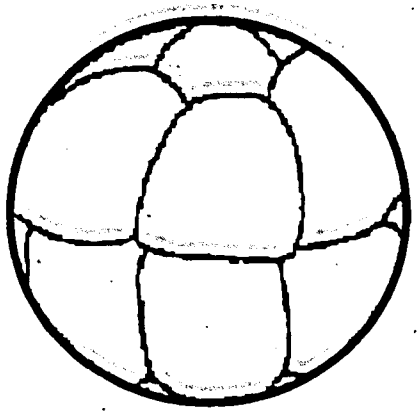
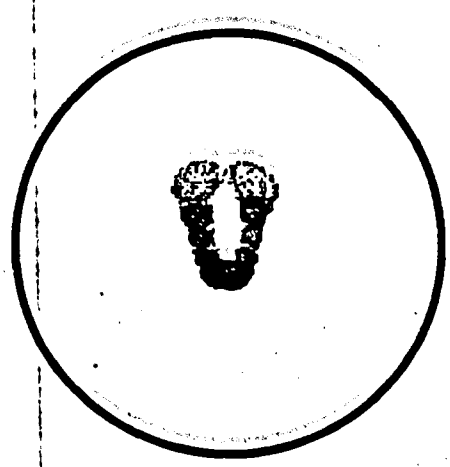


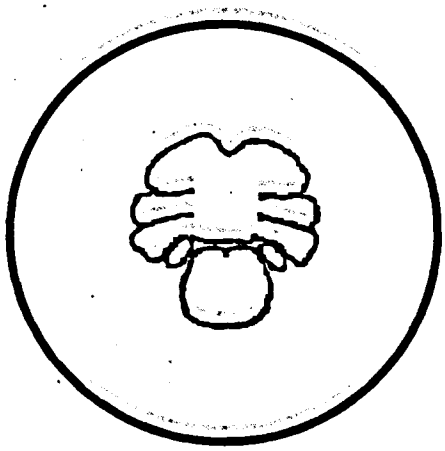
Figure 2 Relationship between gonads colour and gonads stage



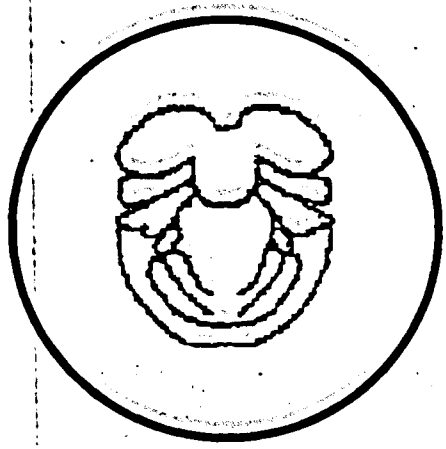
Stage 1 (blastula)



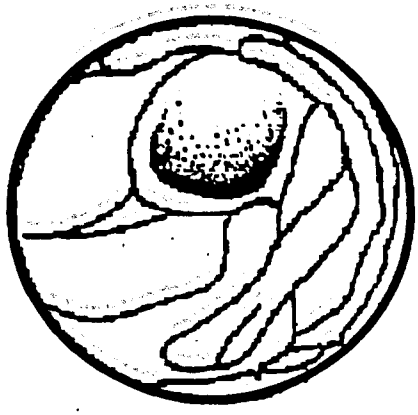
Stage 2 (gastrula)



Stage 3 (nauplius)



Stage 4 (meta-nauplius)



Stage 5 (pigment stage)



Stage 6 (chromatophores formation)

Figure 3 Stages of development of the embryos in eggs incubated under the abdomen of the female (Modified from Kon, 1980)

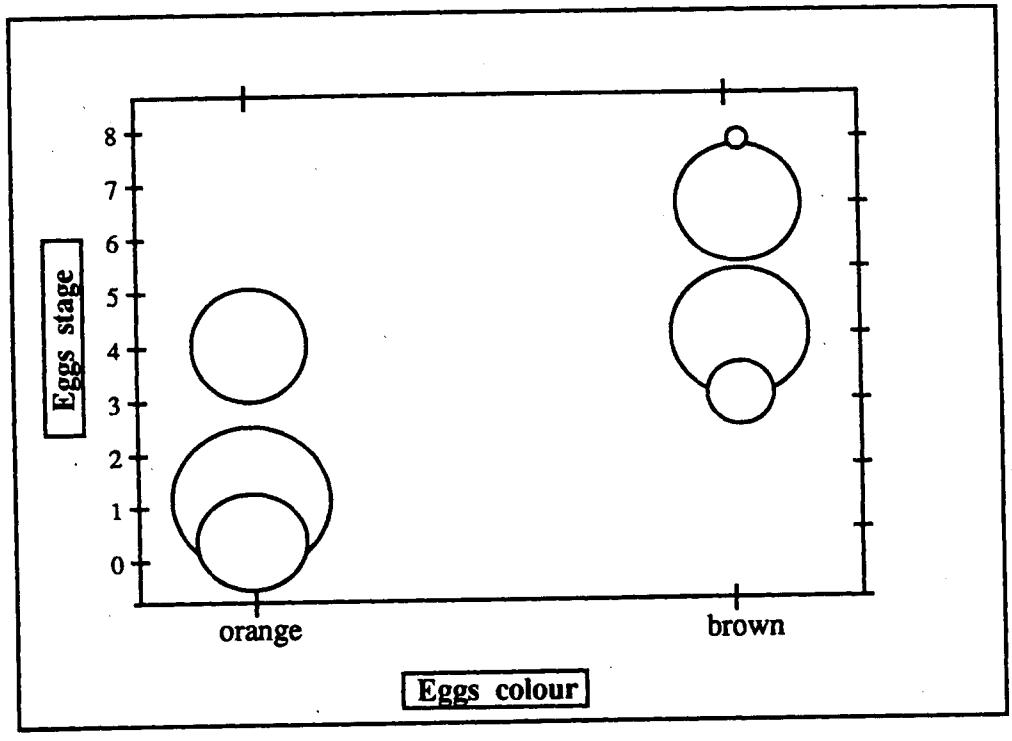


Figure 4 Relationship between eggs colour and eggs development stage

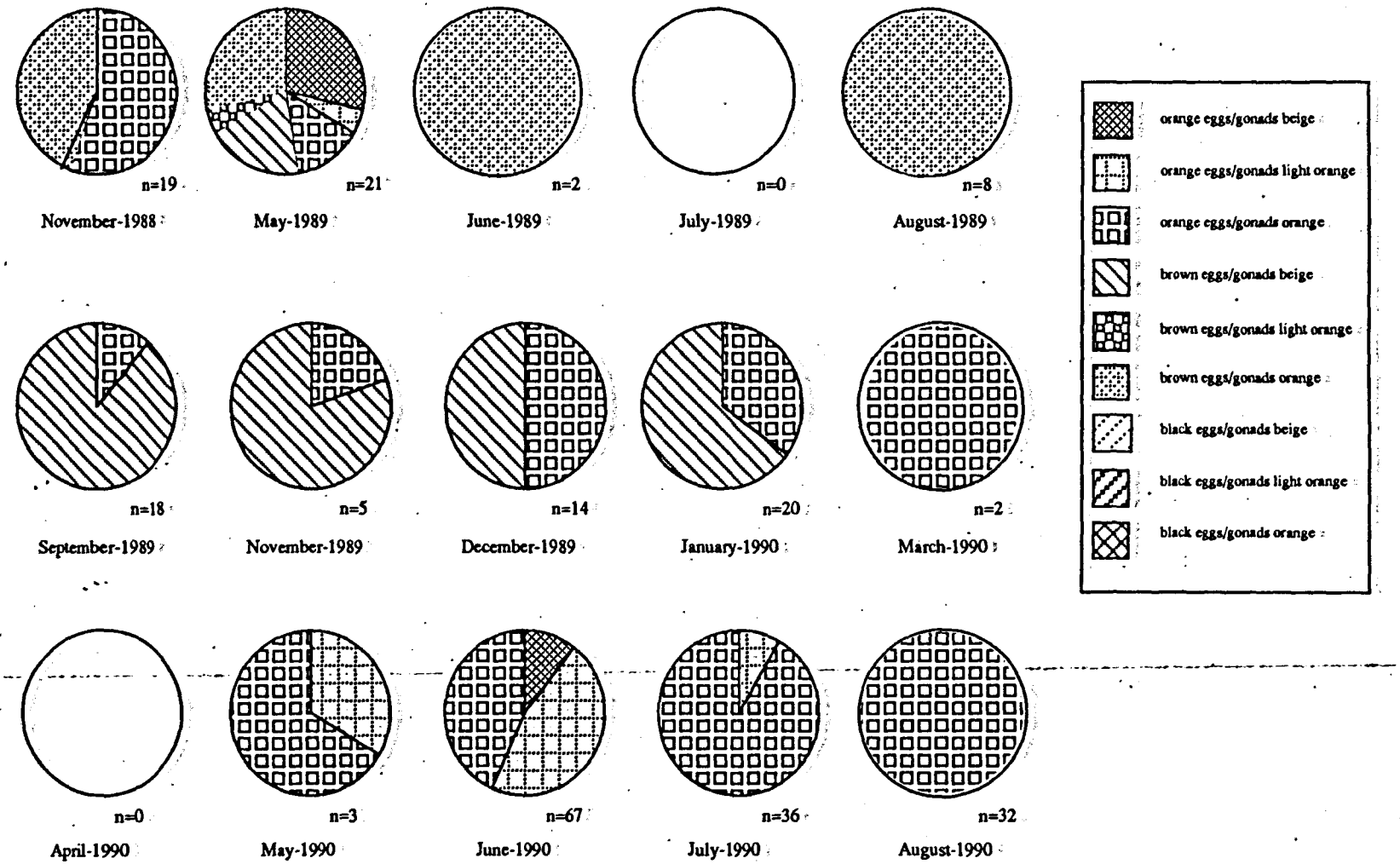


Figure 5a Sample composition for the primiparous female take between november 1988 and August 1990

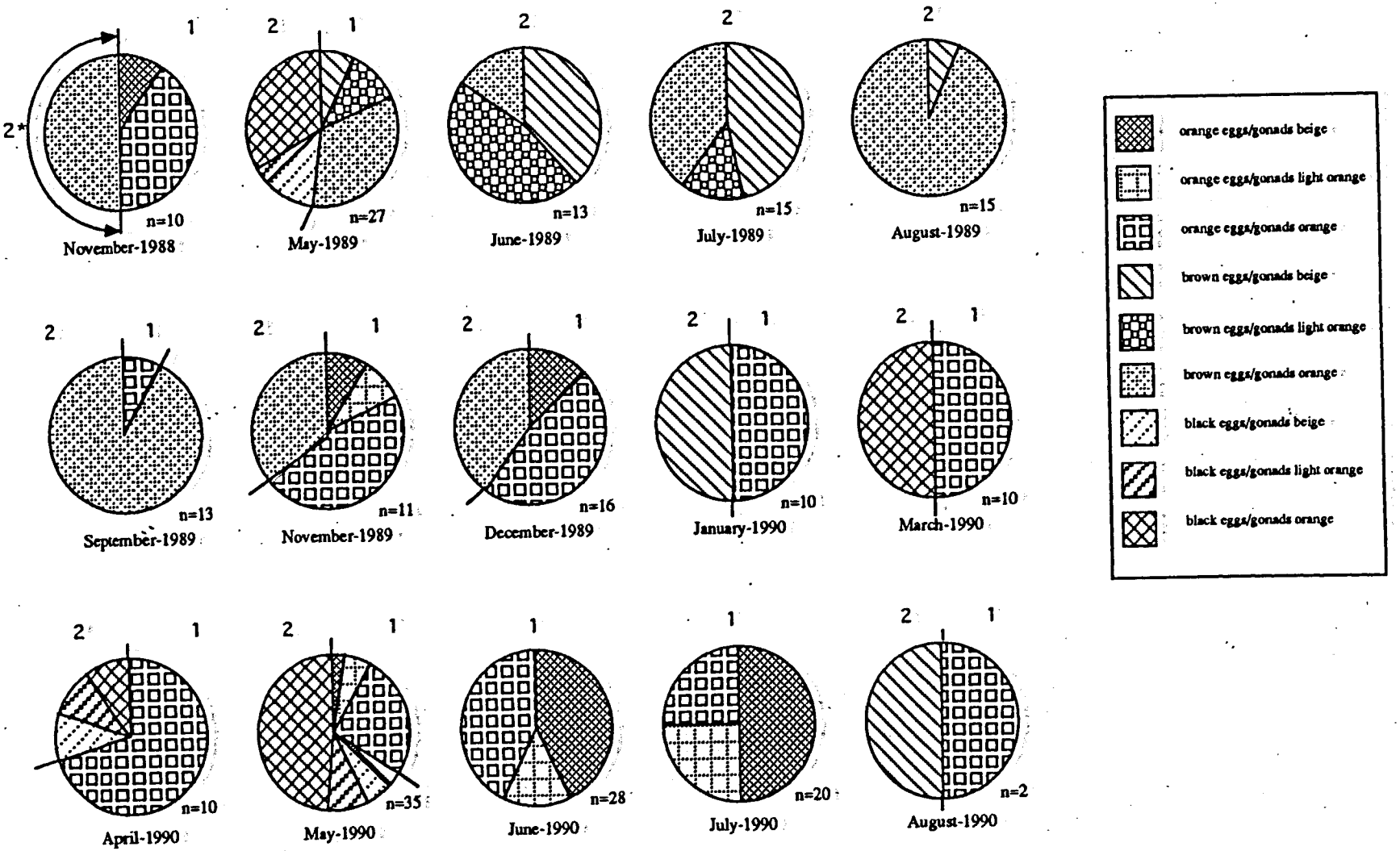


Figure 5b Sample composition for the multiparous female take between november 1988 and August 1990

** first or second year that the female carry the eggs*

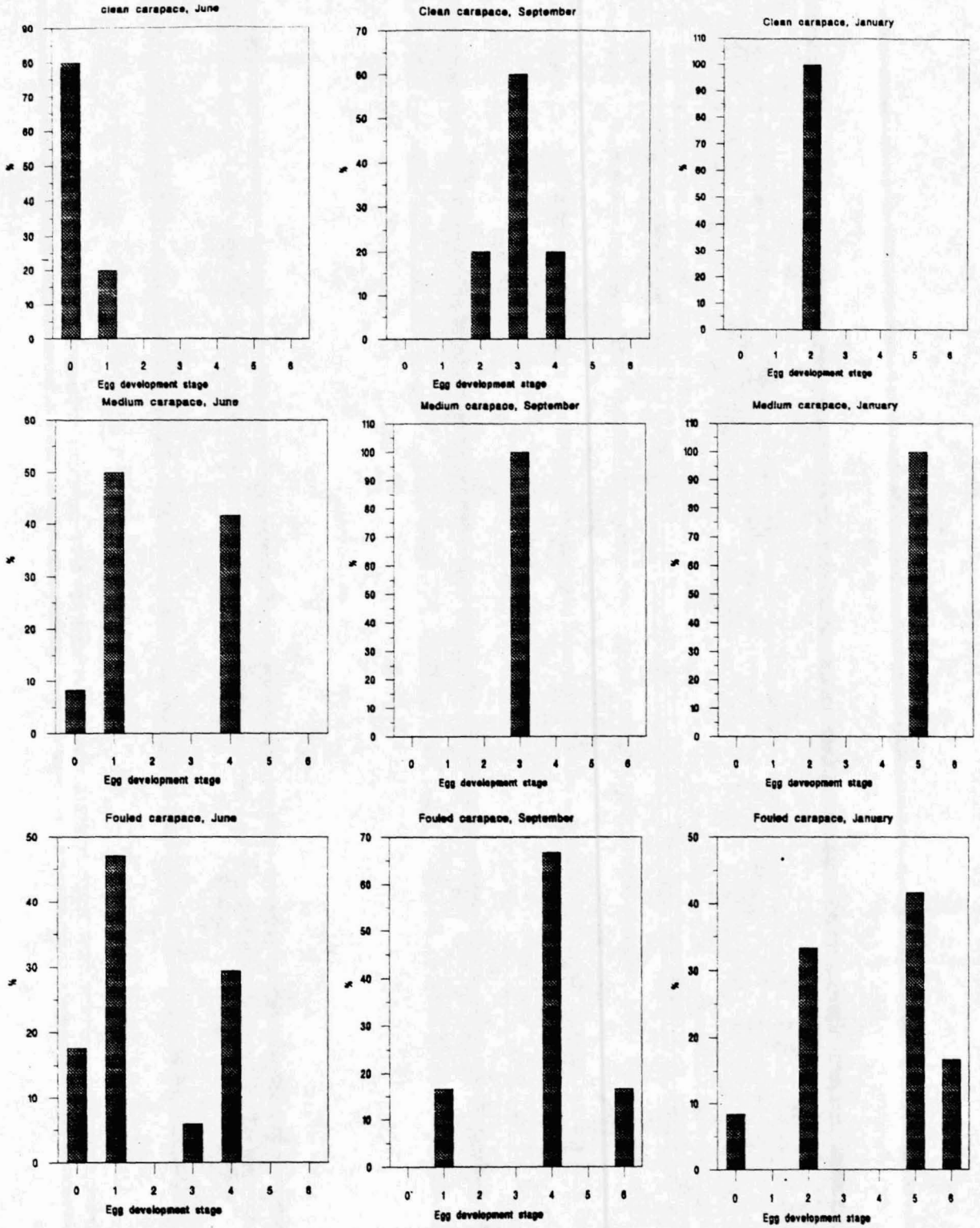


Figure 6a Frequencies of eggs stages in June, September and January for the three categories of females.

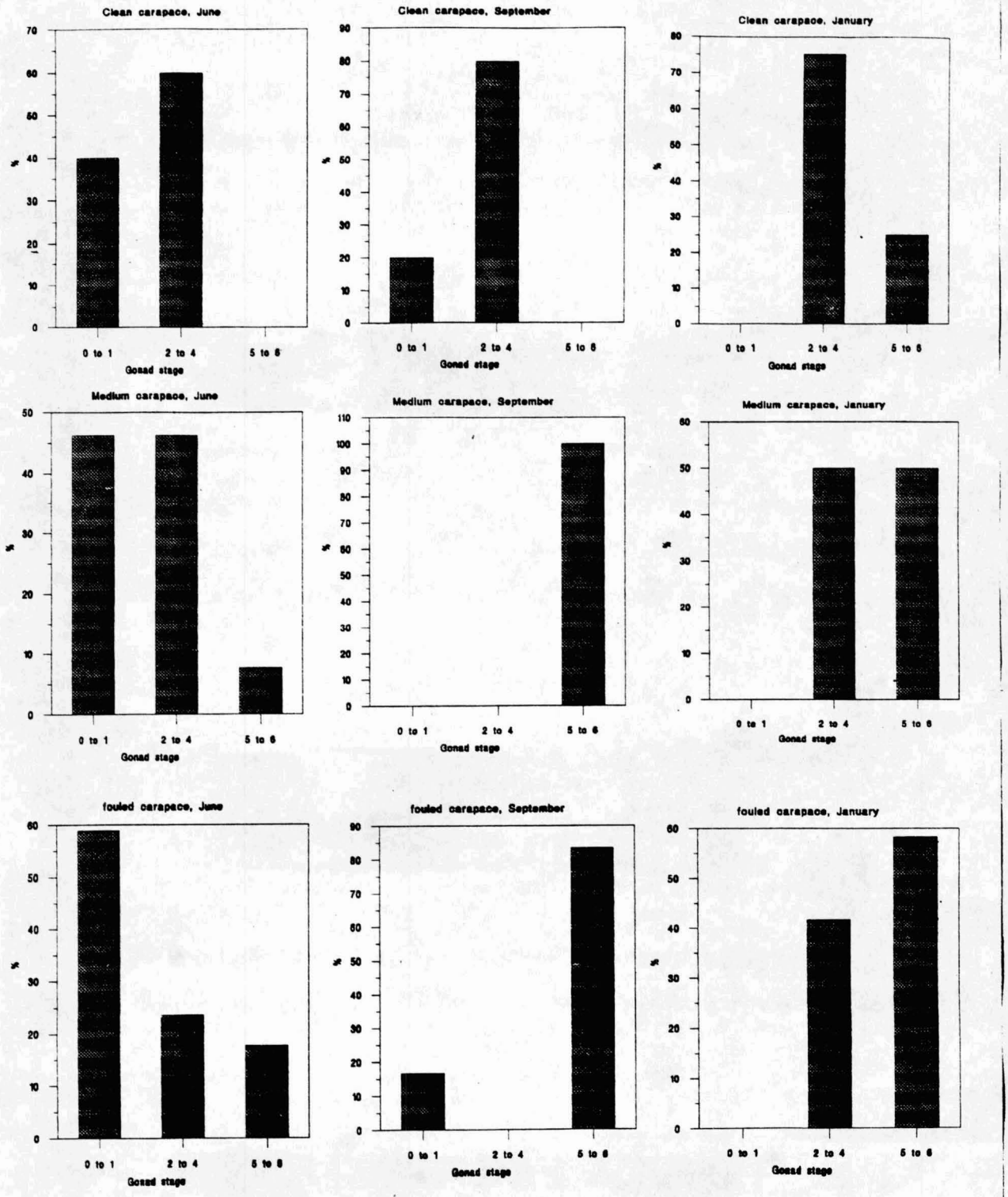


Figure 6b Frequencies of gonad stages in June, September and January for the three categories of females.

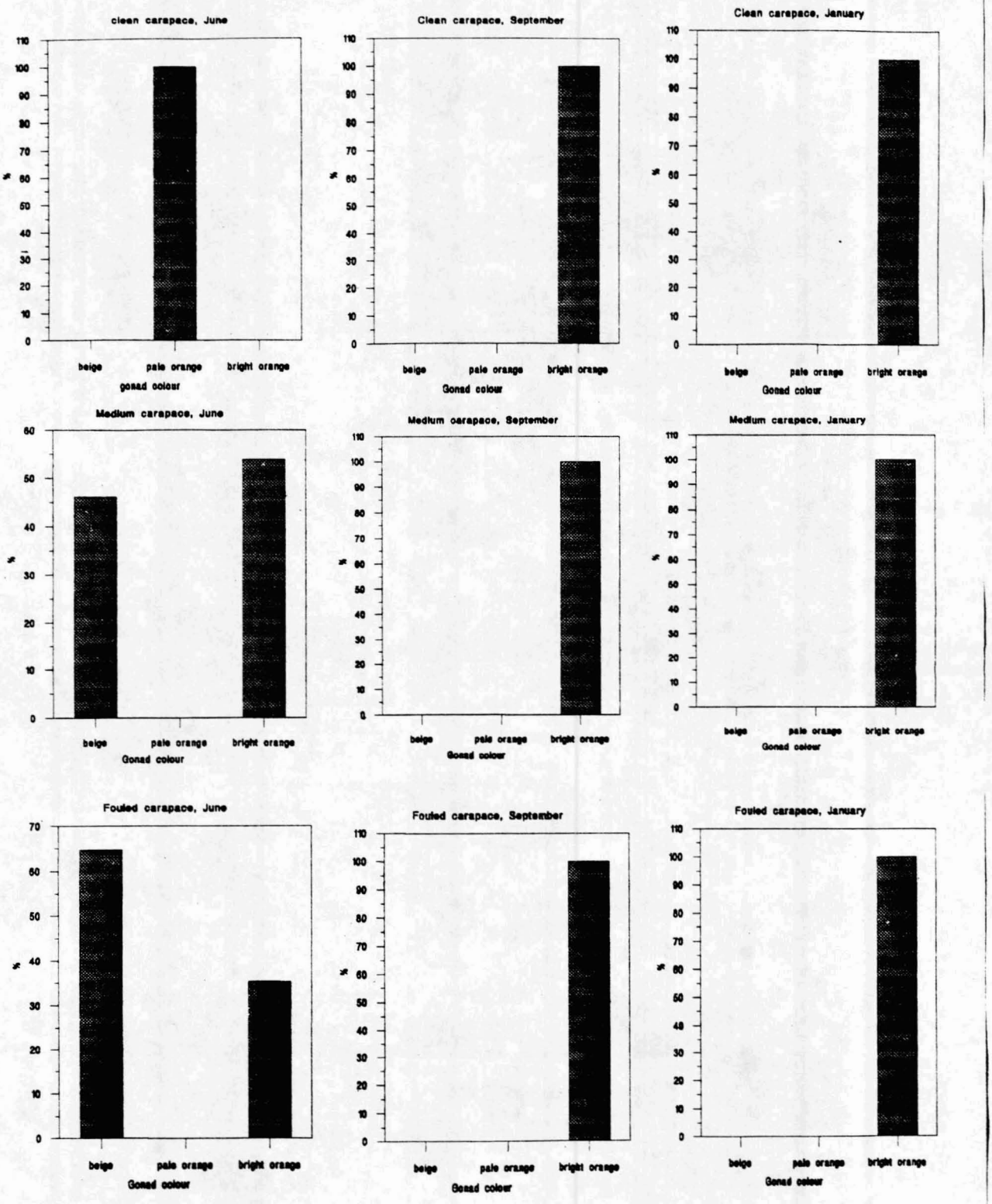


Figure 7a Frequencies of gonad colour in June, September and January for the three categories of females.

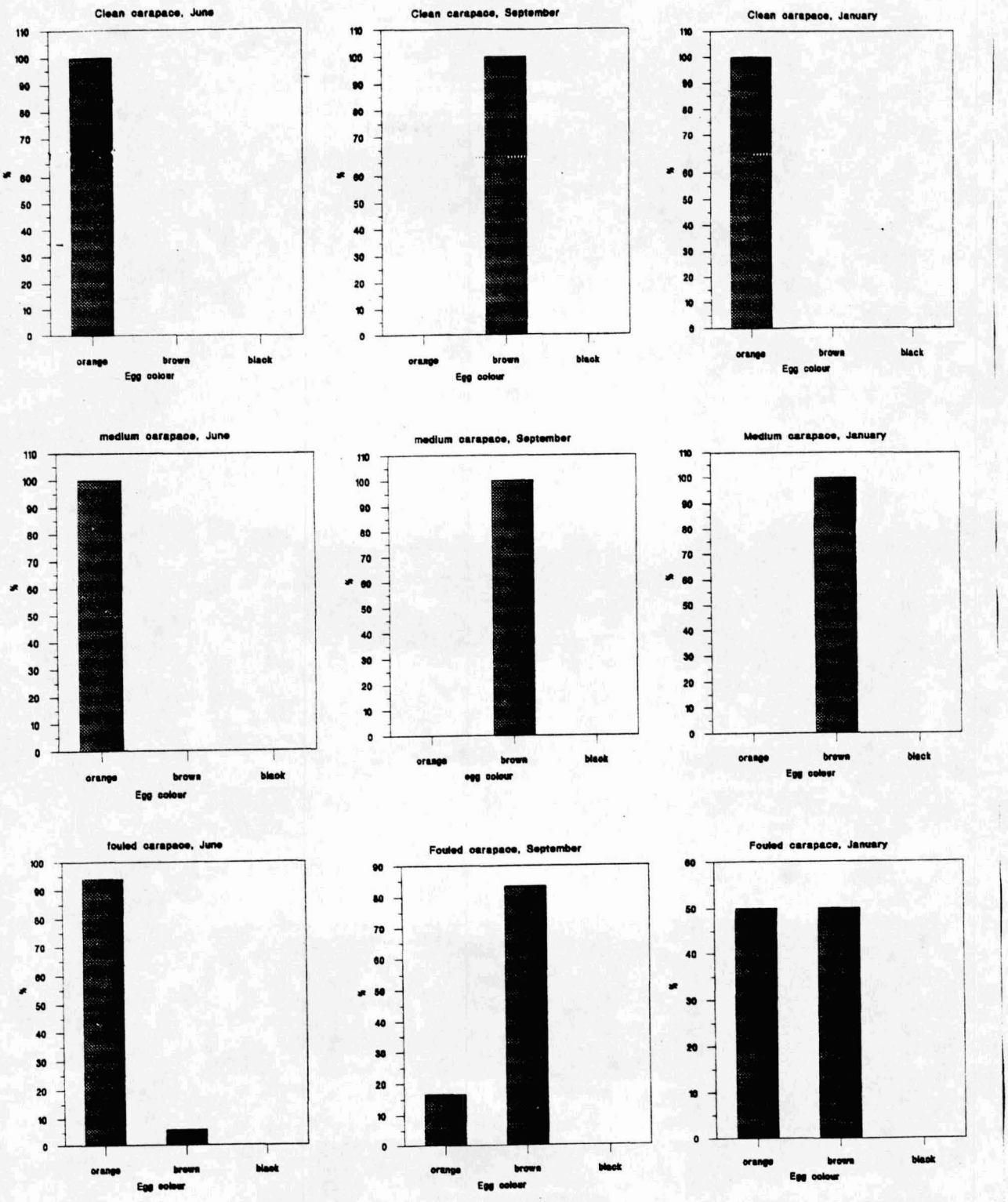


Figure 7b Frequencies of eggs colour in June, September and January for the three categories of females.